

Artificial Intelligence - Complete Guide

1. What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and make decisions.

2. Types of AI

Narrow AI: Designed for specific tasks. General AI: Human-level intelligence. Super AI: Beyond human intelligence.

3. Machine Learning

A subset of AI where machines learn from data. Types: Supervised, Unsupervised, Reinforcement Learning.

4. Deep Learning

Uses neural networks with many layers to model complex patterns in data such as images, text, and speech.

5. Natural Language Processing

Enables machines to understand and process human language. Used in chatbots, translation, sentiment analysis.

6. Computer Vision

Allows machines to interpret visual data such as images and videos.

7. Key Algorithms

Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, SVM, Neural Networks.

8. Applications of AI

Healthcare, Finance, E-commerce, Autonomous Vehicles, Robotics, Education, Cybersecurity.

9. Tools & Libraries

Python, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy.

10. Future of AI

AI is expected to transform industries, create new jobs, and raise ethical challenges.

11. Ethics in AI

Bias, Privacy, Transparency, Accountability are major concerns in AI development.

12. Careers in AI

AI Engineer, Data Scientist, ML Engineer, Research Scientist, NLP Engineer.